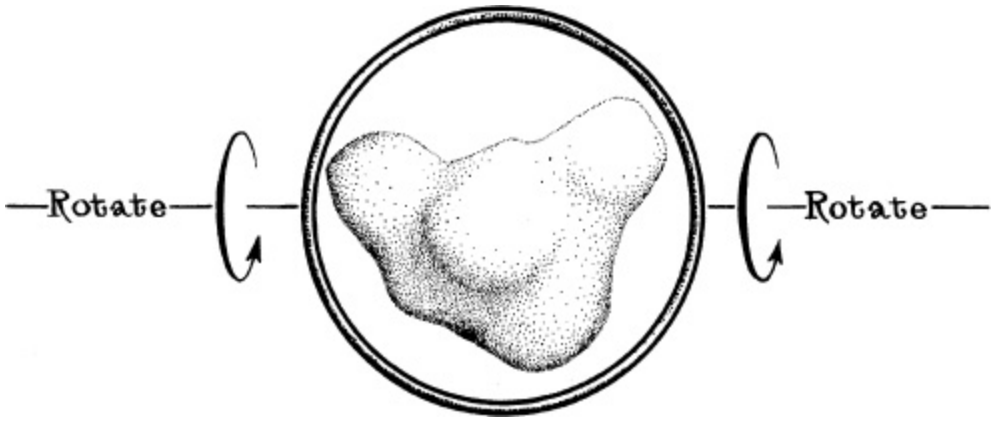
then the object will form a black hole around itself when, and only when, its circumference in all directions has become less than the critical circumference.



7.2 According to the hoop conjecture, an imploding object forms a black hole when, and only when, a hoop with the critical circumference can be placed around the object and rotated.

I proposed this hoop conjecture in 1972. Since then, I and others have tried hard to learn whether it is correct or not. The answer is buried in Einstein's field equation, but to extract the answer has proved exceedingly difficult. In the meantime, circumstantial evidence in favor of the hoop conjecture has continued to mount. Most recently, in 1991, Stuart Shapiro and Saul Teukolsky at Cornell University have simulated, on a supercomputer, the implosion of a highly nonspherical star and have seen black holes form around the imploded star precisely when the hoop conjecture predicts it. If a hoop can be slipped over the imploded star and rotated, a black hole forms; if it cannot, there is no black hole. But only a few such stars were simulated and with special nonspherical shapes. We therefore still do not know for certain, nearly a quarter century after I proposed it, whether the hoop conjecture is correct, but it looks promising.

Igor Dmitrievich Novikov in many ways was my Soviet counterpart, just as Yakov Borisovich Zel'dovich was Wheeler's. In 1962, when I was first meeting Wheeler and embarking on my career under his mentorship, Novikov was first meeting Zel'dovich and becoming a member of his research team.

Whereas I had had a simple and supportive early life—born and reared in a large, tightly knit Mormon family[2] in Logan, Utah—Igor Novikov had